

STATEMENT OF WORK

Title: Primary Vibration Calibration System Upgrade

I. BACKGROUND

The Quantum Measurement Division (QMD) at the National Institute of Standards and Technology (NIST) provides the fundamental basis for the measurement of mass, force, and vibration in the United States by maintaining and modernizing primary standards linked to the International System of Units (SI). The Division performs leading-edge research to develop advanced measurement technologies and provides calibration services that support U.S. industrial productivity and ensure equity in trade.

QMD requires an upgrade to its existing Primary Vibration Calibration System to sustain and enhance these capabilities. This acquisition directly supports the Division's mission by modernizing the technical infrastructure necessary to deliver high-accuracy vibration calibration services. The upgraded system will enable uniaxial excitation of sensors across an extended frequency range while minimizing measurement uncertainty and cross-axis motion.

Purpose/Objective: The purpose of this requirement is to procure and integrate the electronics, software, laser vibrometer, and high-frequency, high-performance shaker necessary to perform primary vibration calibration.

The upgraded system shall include all hardware and software required to routinely deliver low-uncertainty vibration calibrations. It shall meet or exceed the performance of the current NIST Primary Vibration Calibration System (PVCS) across its full operating range and provide the specific performance enhancements identified in the technical specifications below.

The system shall maintain compatibility with existing electrodynamic shakers and shall be supported, including maintenance and parts availability, for a minimum of ten (10) years.

II. SCOPE

The Contractor shall upgrade the existing Primary Vibration Calibration System, which consists of Spektra CS18 vibration calibration system, Polytec OFV-5000/OFV-534 laser vibrometer, Spektra APS129, APS500, and APS600 shakers (with associated power amplifiers and zero-position controllers), and Spektra FECB electrical calibration unit.

III. MINIMUM REQUIREMENTS

The Contractor shall provide a system upgrade that meets all technical specifications identified below.

Line Item 0001:

Description: Vibration Calibration System Upgrade
Quantity: 1

A. Technical Specifications:

1. The system shall provide for primary and secondary calibration of ICP (IEPE), piezoresistive, and capacitive accelerometers, as well as laser vibrometers.
2. The system shall include all data acquisition and processing hardware and software necessary to provide primary vibration calibration when used in conjunction with compatible shakers and laser interferometers / vibrometers.
3. The system shall be compatible with existing components: Spektra APS129 (with power amplifier and zero-position controller), Spektra APS500 (with power amplifier and zero-position controller), and Spektra APS600 (with power amplifier and zero-position controller).
4. The system shall operate up to a frequency of 50 kHz.
5. The system shall include a shaker (with power amplifier and zero-position controller) that performs uniaxial sensor sinusoidal excitation from 5 Hz to 50 kHz.
6. The system shall include a reference laser interferometer / vibrometer for use as a primary calibration reference.
7. The system shall include any required system calibration unit.
8. The shaker shall have a mass load capacity equal to or greater than 2 kg.
9. The vibration calibration system shall accommodate 4 parallel synchronously sampled ICP accelerometer inputs.
10. The laser interferometer / vibrometer optical input/output shall be moveable within a 2.5 meter radius, so as to use with different shakers.
11. The laser interferometer / vibrometer shall have a maximum measurable velocity of not less than 2 m/s.
12. The system, comprising the upgrade and the compatible existing components, shall meet or exceed the performance of the present NIST primary vibration calibration system, in terms of uncertainty provided over the frequency range 100 Hz - 20 kHz for calibration of accelerometers of mass up to 100 g, and in terms of uncertainty provided over the frequency range 0.1 Hz - 100 Hz for calibration of accelerometers of mass up to 1 kg.
13. The system shall be upgradeable in the future for improved performance and functionality.
14. The system shall be supported by the Contractor for a minimum of 10 years.

15. A 24-month warranty on the system components (excluding system calibration unit, if applicable), covering all expenses to replace any defective component, shall be provided.

IV. PLACE OF PERFORMANCE

Installation and demonstration shall be done at 100 Bureau Drive, Gaithersburg, MD 20899.

V. PERIOD OF PERFORMANCE / LEAD TIME

The delivery lead time shall be within twenty (20) weeks after receipt of order. A two (2) year warranty shall be included at no additional cost to the Government, as described in section X. WARRANTY.

NIST's normal business hours: 8:30am to 5:00pm Eastern Time, Monday through Friday except Federal Holidays.

VI. DELIVERY TERMS

Delivery shall be F.O.B Destination. The Contractor shall pack and mark the shipment in conformance with carrier's requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of or damage to the goods occurring before receipt and acceptance of the shipment by the NIST consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery. The Contractor shall coordinate this with the NIST subject matter expert (SME) prior to shipping. Coordination includes shipping method, anticipated delivery date, tracking number, and any special considerations.

The delivery address is:

NIST - Shipping and Receiving (Div 684.07)

(include contract number)
100 Bureau Drive, Building 301
Gaithersburg, MD 20899

Due to ongoing limited site access of the NIST facilities, the Awardee SHALL confirm with the Designated Government Official and the Contracting Officer prior to ANY shipment to NIST

VII. INSTALLATION

The system shall be installed by the Contractor no later than 1 month after delivery. At a minimum, installation shall include unpackaging of all equipment, assembly, demonstration of all manufacturer's specifications, and removal of trash. Installation shall be coordinated with the NIST SME or COR and take place during normal business hours.

VIII. TRAINING

Training in use of all parts of the system shall be provided. The Contractor shall conduct a training session for about 3 persons. The training shall provide a thorough demonstration of all functions, maintenance, calibration, data analysis, and basic troubleshooting. Training shall be within 1 week of installation completion. Training shall be during normal business hours and shall be coordinated with the NIST SME.

IX. INSPECTION & ACCEPTANCE

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance.

Upon delivery, a visual inspection will be performed by the NIST SME to identify defects, damage, or nonconformity. The Government shall have sole discretion to reject, require repair, or require replacement of such items at no cost to the Government. Ownership of the equipment shall transfer to NIST upon acceptance by NIST. Acceptance will be determined after the Contractor has successfully provided the delivery, installation, demonstration, and training. NIST will then conduct testing and inspection for acceptance. Confirmation of acceptance will be provided by the NIST SME within 2 weeks. The Contractor shall assist in troubleshooting and resolving any issues detected during this time within 3 business days after being notified.

Performance Tests:

- The system must be verified to operate up to 50 kHz.
- The reference laser vibrometer must demonstrate a measurable velocity of no less than 2 m/s.
- The parasitic motions of the shaker, and the system optical and electrical measurements, must provide performance meeting or exceeding the system currently installed at NIST in terms of the accuracy of calibration of a 100 g accelerometer from 100 Hz to 20 kHz.
- Synchronous sampling of 4 ICP accelerometer inputs must be demonstrated.

X. WARRANTY

The Contractor shall provide its standard commercial warranty for all products provided under this contract. The Contractor shall extend to the Government the full coverage of any standard commercial warranty normally offered to the public at no additional cost. The Contractor shall warranty the deliverables for 2 years. The warranty period shall commence on the date of acceptance.

XI. NIST SITE ACCESS

Security: NIST is a restricted campus. A site access badge is required for campus and building access and must always be visible. Contractor personnel must comply with NIST identification and access requirements. At a minimum, each Contractor personnel must provide a government

ID (active, unexpired, REAL ID Act compliant) to get a site access badge. The Contractor shall coordinate site access with the NIST SME at least 2 weeks prior to arrival (longer if Contractor personnel include non-US-citizens).

Safety: Contractor personnel are responsible for knowing and complying with all applicable safety regulations. The Contractor shall coordinate this with the NIST SME.

Vehicle Registration: All vehicles entering the site must be registered with the NIST Security Office. A valid driver's license (REAL ID compliant: <https://www.tsa.gov/realid>), active proof of insurance, and current registration must be presented to the NIST Security Office for a vehicle pass. The pass must be displayed on the vehicle in accordance with NIST Security Office instructions.

XII. PAYMENT SCHEDULE

This is a firm-fixed price purchase order. The contractor shall be paid upon acceptance of deliverable and submission of a proper invoice. Payment shall be made in accordance with the Prompt Payment Act. Advance payment is not authorized.